

## Tesamorelin + Ipamorelin **For Research Use Only**

(Tesa 0.5 - 2 mg, + Ipa 100 - 250 mcg 3 – 7 days / week, fasted, subq)

Tesamorelin is a growth hormone-releasing hormone (GHRH) analog that reduces visceral fat and fatty liver disease. Ipamorelin is a selective ghrelin-receptor agonist that produces a discrete growth-hormone pulse when administered intravenously in humans. Because the two peptides stimulate growth-hormone release through complementary receptors, combining them is likely to increase endogenous growth hormone and insulin-like growth factor 1. Community reports describe the combination as producing a stronger, smoother, and more physiologic-feeling response, with improved recovery, body composition, and tolerability.

 Short-term safety is likely. Long-term safety is plausible.

★ ★ The combination is likely to increase GH and IGF-1 because both components independently do so through complementary receptors. This plausibly creates benefits for recovery, fitness, and generalized body recomposition.

★ ★ ★ Reduction in visceral fat is established for Tesamorelin and very likely for the combination.

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

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### Dosage & Patterns

The pivotal phase 3 tesamorelin trials used 2 mg subcutaneously once daily for 26 weeks to reduce visceral fat in adults with HIV-associated abdominal fat accumulation.

The following speculative dosing approaches are commonly discussed:

- **Tesamorelin 2 mg with ipamorelin 250 mcg.**
- **Tesamorelin 0.5 - 1 mg with ipamorelin 100 - 250 mcg**, often selected in community practice in an effort to improve tolerability.
- Reported schedules include **once daily, five days per week**, or **every other day**.
- Cycling: **A 12-weeks ON / 6-weeks OFF pattern** is commonly used in community protocols, and longer cycles such as 12–24 weeks ON followed by 4–12 weeks OFF are also reported. These cycling schedules are speculative and have not been established in controlled trials. Tesamorelin studies generally used continuous daily treatment, including trials extending to 52 weeks. Therefore, laboratory findings, clinical response, and adverse effects provide a

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more biologically defensible basis for determining treatment duration and breaks than an arbitrary fixed cycle.

**Persistent IGF-1 elevation above +3 SDS, clinically significant deterioration in glucose control, or GH/IGF-1–related adverse effects such as edema, arthralgia, or carpal-tunnel symptoms are plausible reasons to discontinue or reduce GH-axis stimulation.** Restarting after IGF-1 has returned to the age-adjusted reference range, glucose abnormalities have resolved or are adequately controlled, and clinically significant adverse effects have resolved is a conservative and biologically plausible approach.

Cycling GH-axis peptides is common in community practice, but the available tesamorelin evidence does not demonstrate that cycling is necessary to maintain responsiveness. In controlled trials, reductions in visceral adipose tissue were maintained with continuous tesamorelin treatment for up to 52 weeks. Anti-tesamorelin antibodies were common, but mean VAT reduction and IGF-1 responses were similar in participants with and without antibodies, providing no clear evidence that treatment breaks are required to prevent loss of efficacy.

For a finite 26- or 52-week course, therefore, there is no clinical evidence that scheduled cycling is necessary if laboratory values and tolerability remain acceptable. For indefinite or repeated long-term use periodic OFF periods can nevertheless be justified as a conservative, mechanistically plausible strategy to reduce cumulative GH/IGF-1 exposure.

### **Benefits**

Tesamorelin has demonstrated human benefits for visceral fat, liver fat, lipid profiles, lean body mass, muscle composition, and cognition in the populations studied. Ipamorelin is justified mechanistically to likely increase the endogenous GH pulse and benefits.

A 12-week randomized, placebo-controlled trial in adults with type 2 diabetes compared tesamorelin 1 mg and 2 mg subcutaneously once daily. Both doses significantly increased IGF-1 relative to placebo, with mean increases of 33 ng/mL at 1 mg and 66 ng/mL at 2 mg. Insulin response, fasting glucose, overall diabetes control, and body-composition measures remained similar among the groups over 12 weeks. It is likely these benefits will apply to the general population.

Injected ipamorelin produces a discrete, concentration-dependent GH pulse in humans.

- **Visceral fat reduction:** In adults with HIV-associated abdominal fat accumulation, tesamorelin 2 mg subcutaneously once daily produced a pooled placebo-adjusted 15.4% reduction in visceral adipose tissue at 26 weeks. It also reduced waist circumference and waist-to-hip ratio while largely preserving abdominal subcutaneous fat.

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- **Improved lipid profile:** In pooled phase 3 trials involving adults with HIV-associated abdominal fat accumulation, tesamorelin reduced triglycerides and improved the ratio of total cholesterol to high-density lipoprotein cholesterol.
- **Reduced liver fat:** In a 12-month randomized trial involving people with HIV and nonalcoholic fatty liver disease, tesamorelin reduced liver fat and reduced progression of liver fibrosis.
- **Improved executive function:** In a 20-week randomized trial involving healthy older adults and older adults with mild cognitive impairment, tesamorelin 1 mg daily improved overall cognition and executive function. Delayed verbal recall also improved in the subgroup with mild cognitive impairment.
- **Increased lean body mass:** In a 12-week randomized dose-ranging trial involving adults with HIV-associated abdominal fat accumulation, tesamorelin 1 mg and 2 mg daily increased lean body mass relative to placebo.
- **Improved muscle composition:** Among adults with HIV-associated abdominal fat accumulation who responded to tesamorelin, trunk-muscle density and lean muscle increased over 26 weeks.

### Potential Applications

- **Excess visceral fat:** Tesamorelin reduces visceral fat in adults with HIV-associated abdominal fat accumulation and in adults with abdominal obesity and reduced growth-hormone secretion.
- **Fatty liver:** Tesamorelin reduces liver fat and fibrosis progression in people with HIV and nonalcoholic fatty liver disease.
- **Cognitive, recovery, fitness, and generalized body recomposition through GH-axis support:** Tesamorelin's benefits in these areas are plausibly increased by adding ipamorelin through complementary receptor activity.

### Cautions

- **Previous malignancy:** A preexisting malignancy should be inactive and its treatment complete before tesamorelin is considered.
- **Pregnancy or Breastfeeding:** Tesamorelin should be avoided during pregnancy or breastfeeding.
- **Diabetes or glucose intolerance:** Tesamorelin can cause glucose intolerance or diabetes, so glucose status should be assessed before and during use.
- **Persistently elevated IGF-1:** Reassess use when insulin-like growth factor 1 remains above the age-adjusted range. Discontinuation should be considered when it remains more than three standard deviations above the age-adjusted mean.
- **Concerning growth-related symptoms:** Unexplained masses, rapid tissue growth, or other concerning symptoms warrant clinical assessment and reassessment of continued use.

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### Side Effects

- **Fluid retention:** Peripheral edema can occur and is often most apparent early in treatment.
- **Musculoskeletal symptoms:** Joint pain and muscle pain can occur.
- **Headache and nausea:** Both have been reported in clinical studies.
- **Changes in sleep:** Deeper sleep and vivid dreams are reported in peptide communities and remain speculative as combination-specific effects.

### Cancer Risk

Tesamorelin increases circulating insulin-like growth factor 1. After 26 weeks of treatment, 47% of participants had levels more than two standard deviations above the age-adjusted mean and 36% had levels more than three standard deviations above the mean. Insulin-like growth factor 1 participates in cellular proliferation, and higher circulating levels are epidemiologically associated with several cancers. It is therefore plausible that sustained growth-hormone and insulin-like-growth-factor-1 elevation could promote the growth of an occult or recurrent malignancy.

### Drivers of Diminished Response and Mitigation Strategies

Both GHRH receptor agonism and ghrelin receptor stimulation can produce **receptor desensitization** with continuous exposure as described in pituitary physiology literature. Tesamorelin clinical trial reports document **neutralizing antibodies** in some subjects, so antibody mediated loss is a real documented risk. IGF-1 feedback and intracellular signaling adjustments are expected so **downstream pathway adaptation** and reduce efficacy over time is likely. **Overall likelihood of clinically meaningful waning is high.**

Mitigation supported by the evidence is **cycling**.

### Biological Mechanism

Tesamorelin and ipamorelin stimulate pituitary somatotrophs through distinct receptors. Tesamorelin is a stabilized analog of growth-hormone-releasing hormone that activates the growth-hormone-releasing-hormone receptor. Ipamorelin is a selective agonist of growth-hormone-secretagogue receptor 1a. Because both agents independently stimulate growth-hormone release in humans, combining them is likely to increase endogenous growth-hormone output through complementary receptor activity. Preservation of normal endocrine feedback during combined use is mechanistically plausible.

### Receptor Binding and Intracellular Signaling

Tesamorelin activates growth-hormone-releasing-hormone receptors on pituitary somatotrophs. These receptors couple primarily to the stimulatory G protein Gs, increasing adenylyl cyclase activity, cyclic adenosine monophosphate, and protein kinase A signaling that supports growth-

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hormone synthesis and secretion. Ipamorelin activates growth-hormone-secretagogue receptor 1a, which couples primarily to Gq/11 proteins and activates phospholipase C, inositol trisphosphate, and intracellular calcium signaling. These pathways converge on the cellular machinery controlling growth-hormone release, making an additive secretory effect from combined activation mechanistically plausible.

### **Pituitary Dynamics and GH Pulsatility**

Tesamorelin increases basal and pulsatile endogenous growth-hormone secretion in humans. Ipamorelin produces a discrete growth-hormone pulse that peaks rapidly and then declines. Combining the two agents is likely to increase total endogenous growth-hormone output through complementary receptor activity more than either alone with a larger pulse amplitude, modestly extended pulse duration.

### **Peripheral Growth-Hormone and IGF-1 Effects**

Tesamorelin-induced growth-hormone pulses activate hepatic growth-hormone receptors and increase circulating insulin-like growth factor 1. Its effects plausibly involve increased lipolysis and lipid oxidation, altered very-low-density-lipoprotein metabolism, growth-hormone-receptor signaling through Janus kinase 2, signal transducer and activator of transcription 5, and mitogen-activated protein kinase pathways, and insulin-like-growth-factor-1 signaling that supports protein synthesis and mitochondrial function.

### **Pharmacokinetics and Enzymatic Stability**

Tesamorelin contains an amino-terminal trans-3-hexenoyl modification that increases resistance to enzymatic degradation and improves stability relative to native growth-hormone-releasing hormone. In human intravenous studies, ipamorelin had an approximately two-hour terminal half-life and produced a growth-hormone response that peaked at about 40 minutes and declined to low levels within six hours. These distinct properties make a temporally complementary pharmacodynamic profile during combined use plausible.

## **Conclusions**

Tesamorelin increases growth hormone and insulin-like growth factor 1 and has demonstrated human benefits for visceral fat, liver fat, lipid profiles, lean body mass, muscle composition, and cognition in the populations studied. Intravenous ipamorelin produces a discrete growth-hormone pulse, and combining the two agents is likely to increase endogenous growth hormone and insulin-like growth factor 1 through complementary receptor activity. Greater overall efficacy, a smoother pulse pattern, and better tolerability are widely reported in peptide communities but remain speculative as combination-specific effects.

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